where in (*) we use (5) and the assumption $\rho \geq 3d$. For the first term of (4), we know $\hat{A}_i\phi(\hat{x}^{i-1}) = \hat{x}^i_{i-1} + \Delta_i\phi(\hat{x}^{i-1})$. Therefore by the induction hypothesis and (3) for $i = j$,

$$\|\hat{A}_i\phi(\hat{x}^{i-1}) - x^i\| \leq \|\hat{x}^i_{i-1} - x^i\| + \|\Delta_i\phi(\hat{x}^{i-1})\| \leq (i-1)\epsilon'\|x^i\|/d + \epsilon'\|x^i\|/3d \leq \epsilon'\|x^i\|,$$

so again we have

$$\|(M^{i,j} - J^{i,j}_{x^i})(\hat{A}_i\phi(\hat{x}^{i-1}))\| \leq (\epsilon'/3d)\|x^j\|. \tag{7}$$

Together, (7) and (6) show that (4) is $\leq \frac{2\epsilon'}{3d}\|x^j\|$. Together with (3) we obtain from 2 that $\|\hat{x}^j_i - \hat{x}^j_{i-1}\| \leq \frac{\epsilon'}{d}\|x^j\|$, and hence that $\|\hat{x}^j_i - x^j\| \leq \frac{i\epsilon'\|x^j\|}{d}$, completing the induction step.

Conditioning on the success of interlayer smoothness and inequality (1), we've shown,

$$\|\hat{x}^j_i - x^j\| \leq (i/d)\epsilon'\|x^j\|,$$

for any $i \leq j \leq d$. Recall that with probability at least $1/2 - \delta$, interlayer smoothness holds and inequality (1) holds for every $2 \leq i \leq d$. Thus, let $\epsilon' = O\left(\sqrt{\frac{pc^2d^2\log(mdh_d/\delta)}{h_{\min}\min_{2\leq i\leq d}\mu_i^2\mu_{i\to}^2}}\right)$, we know with probability at least $1/2 - \delta$,

$$\|f_\theta(x) - f_{\theta_1}(x)\| = \|x^d - \hat{x}^d_d\| \leq \epsilon'\|f_\theta(x)\|.$$

Bounding $\|f_\theta(x) - f_{\theta_t}(x)\|$ for any fixed t: The proof for a fixed network on the path is almost the same as the proof for the end point. Instead of considering $\hat{x}^j_i$, now we consider $\hat{x}^j_i(t)$, which is the vector before activation at layer j if the weights $A_2, \ldots, A_i$ are replaced by $A_2 + t(\hat{A}_2 - A_2), \ldots, A_i + t(\hat{A}_i - A_i)$. We can still use Lemma 5 to bound the noise produced by replacing the weight matrix at a single layer because

$$\|U(A_i + t(\hat{A}_i - A_i) - A_i)x\| = t\|U(\hat{A}_i - A_i)x\| \leq \|U(\hat{A}_i - A_i)x\|.$$

Thus, we can still use the above induction proof to show that for any fixed $0 \leq t \leq 1$, let $\epsilon' = O\left(\sqrt{\frac{pc^2d^2\log(mdh_d/\delta)}{h_{\min}\min_{2\leq i\leq d}\mu_i^2\mu_{i\to}^2}}\right)$, with probability at least $1/2 - \delta$,

$$\|f_\theta(x) - f_{\theta_t}(x)\| \leq \epsilon'\|f_\theta(x)\|.$$

Bounding $\|f_\theta(x) - f_{\theta_t}(x)\|$ for every t: Finally, we show that $\|f_\theta(x) - f_{\theta_t}(x)\|$ is bounded for every point on the path via an ϵ'-net argument. Similar to previous steps, letting $\epsilon' = O\left(\sqrt{\frac{pc^2d^2\max_{x\in S}(\|f_\theta(x)\|^2)\log(mdh_d/\delta)}{h_{\min}\min_{2\leq i\leq d}(\mu_i^2\mu_{i\to}^2)}}\right)$, we know that with probability at least $1/2 - \delta$,

$$\|f_\theta(x) - f_{\theta_1}(x)\| \leq \epsilon'/2.$$

Next, we show that on the path, the network output is smooth in terms of the parameters. According to Algorithm 1, we know for any $2 \leq i \leq d$, we have $\|\hat{A}_i\| \leq 4\|A_i\|$, so $\|\hat{A}_i - A_i\| \leq 5\|A_i\|$. For any $2 \leq i \leq d$, let $A_{i,t} = A_i + t(\hat{A}_i - A_i)$. Note $\|A_{i,t}\| \leq (1-t)\|\hat{A}_i\| + t\|A_i\| \leq 4\|A_i\|$. For any t, t' and any $2 \leq i \leq d$, let $\theta^i_{t,t'}$ be θ_t with the weight matrix at every layer $2 \leq j \leq i$ replaced by $(A^j + t'(\hat{A}^j - A^j))$. For convenience, we also denote θ_t as $\theta^1_{t,t'}$. Given $\tau < 1/2$, for any $\tau \leq t \leq 1 - \tau$ and for any $-\tau \leq \kappa \leq \tau$, we can bound $\|f_{\theta_{t+\kappa}}(x) - f_{\theta_t}(x)\|$ as follows:

$$\|f_{\theta_{t+\kappa}}(x) - f_{\theta_t}(x)\| \leq \sum_{2\leq i\leq d}\|f_{\theta^i_{t,t+\kappa}}(x) - f_{\theta^{i-1}_{t,t+\kappa}}(x)\|$$

The output of layer $i-1$ is the same for the two networks, of norm $\leq \|x\| \prod_{j=1}^{i-1} \|A_{j,t+\kappa}\|$. Hence the output of layer i differs by at most $\kappa\|x\|\|\hat{A}_i - A_i\| \prod_{j=1}^{i-1} \|A_{j,t+\kappa}\|$ and the output differs by $\kappa\|x\|\|\hat{A}_i - A_i\| \prod_{j=1}^{i-1} \|A_{j,t+\kappa}\| \prod_{j=i+1}^{d} \|A_{j,t}\| \leq 5^d \kappa \|x\| \prod_{j=1}^{d} \|A_j\|$. Hence

$$\begin{aligned}\|f_{\theta_{t+\alpha}}(x) - f_{\theta_t}(x)\| &\leq \sum_{2\leq i\leq d} 5^d \|x\|\kappa \prod_{1\leq j\leq d} \|A_j\| \\ &\leq 5^d d\kappa\|x\| \prod_{1\leq j\leq d} \|A_i\|.\end{aligned}$$

Thus, given $\tau \leq \frac{\epsilon'}{2\cdot 5^d d \max_{x\in S}\|x\| \prod_{1\leq j\leq d}\|A_j\|}$, we know for any $\tau \leq t \leq 1-\tau$ and for any $-\tau \leq \alpha \leq \tau$,

$$\|f_{\theta_{t+\alpha}}(x) - f_\theta(x)\| \leq \epsilon'/2. \tag{8}$$

There exists a set $Q = \{\theta_t\}$ with size $O(1/\tau)$ such that for any network on the path, the distance to the closest network in Q is no more than τ. If we can prove for any $\theta_t \in Q$, $\|f_\theta(x) - f_{\theta_t}(x)\| \leq \epsilon'/2$, we immediately know for any network $\theta_{t'}$ on the path $\|f_\theta(x) - f_{\theta_{t'}}(x)\| \leq \epsilon'$ by inequality (8).

By a union bound over Q, letting $\epsilon' = O\left(\sqrt{\frac{pc^2d^2 \max_{x\in S}(\|f_\theta(x)\|^2)\log\left(\frac{mdh_d}{\delta\tau}\right)}{h_{\min}\min_{2\leq i\leq d}(\mu_i^2\mu_{i\to}^2)}}\right)$, we know with probability at least $1/2-\delta$,

$$\|f_\theta(x) - f_{\theta_t}(x)\| \leq \epsilon'/2,$$

for any $\theta_t \in Q$.

Setting $\delta = 1/4$, we know there exists

$$\epsilon' = O\left(\sqrt{\frac{pc^2d^3 \max_{x\in S}(\|f_\theta(x)\|^2)\log\left(\frac{mdh_d \max_{x\in S}\|x\| \prod_{1\leq j\leq d}\|A_j\|}{\epsilon'}\right)}{h_{\min}\min_{2\leq i\leq d}(\mu_i^2\mu_{i\to}^2)}}\right)$$

such that with probability at least $1/4$,

$$\|f_\theta(x) - f_{\theta_t}(x)\| \leq \epsilon'$$

for any $x \in S$ and any $0 \leq t \leq 1$. Since the loss function is β-Lipschitz, we further know that for any $0 \leq t \leq 1$:

$$L(f_{\theta_t}) \leq L(f_\theta) + \beta\epsilon' = L(f_\theta) + \widetilde{O}(\sqrt{p}\epsilon).$$

□

Now, we are ready to prove the main theorem.

Theorem 2. *Let θ^A and θ^B be two fully connected networks that are both ϵ-noise stable, there exists a path with 10 line segments in parameter space $\pi : [0,1] \to \Theta$ between θ^A and θ^B such that*[6] $L(f_{\pi(t)}) \leq \max\{L(f_{\theta^A}), L(f_{\theta^B})\} + \widetilde{O}(\epsilon)$ *for* $0 \leq t \leq 1$.

Proof of Theorem 2. Setting dropout probability $p = 3/4$, by Lemma 5 and Lemma 3, if $h_{\min} = \widetilde{\Omega}(1)$, we know there exist θ_1^A and θ_1^B such that

1. in both networks, each weight matrix from layer 2 to layer d has at least half of columns as zero vectors;
2. $L(f_{\theta_t^A}) \leq L(f_{\theta^A}) + \widetilde{O}(\epsilon)$ and $L(f_{\theta_t^B}) \leq L(f_{\theta^B}) + \widetilde{O}(\epsilon)$, for any $0 \leq t \leq 1$, where $\theta_t^A = \theta^A + t(\theta_1^A - \theta^A)$ and $\theta_t^B = \theta^B + t(\theta_1^B - \theta^B)$.

[6] Here $\widetilde{O}(\cdot)$ hides log factors on relevant factors including $|S|, d, \|x\|, 1/\epsilon$ and $h_i\|A_i\|$ for layers $i \in [d]$.

Since the dropout fraction in both θ_1^A and θ_1^B is at least half, we can connect θ_1^A and θ_1^B as we did in Lemma 2, while ensuring the loss doesn't exceed $\max\{L(f_{\theta^A}), L(f_{\theta^B})\} + \widetilde{O}(\epsilon)$. Connecting θ^A to θ_1^A and connecting θ_B to θ_1^B each take one line segment. By the construction in Lemma 2, connecting two dropped-out networks θ_1^A and θ_1^B takes 8 line segments. Thus, overall the path between θ^A and θ^B contains 10 line segments. □

Next, we show that if there exists a "narrow" neural network achiving small loss, we can get a lower energy barrier using a smaller dropout probability.

Theorem 3. *Suppose there exists a network θ^* with layer width h_i^* for each layer i that achieves loss $L(f_{\theta^*})$, and minimum hidden layer width $h_{min}^* = \widetilde{\Omega}(1)$. Let θ^A and θ^B be two ϵ-noise stable networks. For any dropout probability $1.5 \max_{1\le i\le d-1}(h_i^*/h_i) \le p \le 3/4$, if for any $2 \le i \le d$, $1 \le j \le h_{i-1}$, $\|[A_i]_j\| = O(\sqrt{p})\|A_i\|_F$ then there exists a path with 13 line segments in parameter space $\pi : [0,1] \to \Theta$ between θ^A and θ^B such that $L(f_{\pi(t)}) \le \max\{L(f_{\theta^A}) + \widetilde{O}(\sqrt{p}\epsilon), L(f_{\theta^B}) + \widetilde{O}(\sqrt{p}\epsilon), L(f_{\theta^*})\}$ for $0 \le t \le 1$.*

Proof of Theorem 3. Since $h_{\min} \cdot \max_{1\le i\le d-1}(h_i^*/h_i) \ge h_{\min}^* = \widetilde{\Omega}(1)$, we have $h_{min} = \widetilde{\Omega}\left(\frac{1}{\max_{1\le i\le d-1}(h_i^*/h_i)}\right)$. By Lemma 5 and Lemma 3, there exist θ_1^A and θ_1^B such that

1. in both networks, each weight matrix from layer 2 to layer d has at least h_i^* columns set to zero;
2. $L(f_{\theta_t^A}) \le L(f_{\theta^A}) + \widetilde{O}(\sqrt{p}\epsilon)$ and $L(f_{\theta_t^B}) \le L(f_{\theta^B}) + \widetilde{O}(\sqrt{p}\epsilon)$, for any $0 \le t \le 1$, where $\theta_t^A = \theta^A + t(\theta_1^A - \theta^A)$ and $\theta_t^B = \theta^B + t(\theta_1^B - \theta^B)$.

From the fact that at least h_i^* units in layer i of both θ_1^A and θ_1^B have been set to zero for $1 \le i < d$—meaning that the corresponding rows of A_i and columns of A_{i+1} are zero—it follows from Lemma 2 that we can connect θ_1^A to an arbitrary permutation of θ^* using 8 segments while keeping the loss on the path no more than $\max\{L(f_{\theta_1^A}), L(f_{\theta^*})\}$. By choosing this permutation so that the non-zero units of θ^* do not intersect with those of θ_1^B, we can then connect θ^* to θ_1^B using just 3 segments as done in the first step of our path construction in Lemma 2 seeing as there is no need to permute θ^* a second time. Combining these paths together with the paths that interpolate between the original parameters θ^A and θ^B and their dropout versions θ_1^A and θ_1^B, we obtain a path in parameter space $\pi : [0,1] \to \Theta$ between θ^A and θ^B with 13 line segments such that $L(f_{\pi(t)}) \le \max\{L(f_{\theta^A}) + \widetilde{O}(\sqrt{p}\epsilon), L(f_{\theta^B}) + \widetilde{O}(\sqrt{p}\epsilon), L(f_{\theta^*})\}$ for $0 \le t \le 1$. □

C Proofs for disconnected modes in two-layer nets

Proof of Theorem 4. Define our loss over parameter space such that $L(f_\theta) = \frac{1}{n}\sum l(y_i, f_\theta(\mathbf{x}_i))$, where $\mathbf{x}_i \in \mathbb{R}^{h+2}$ is our i^{th} data sample, $y_i \in \mathbb{R}$ the associated label, and $f_\theta(\mathbf{x}_i) = \mathbf{w}^T\phi(A\mathbf{x}_i)$ for $\theta = (\mathbf{w}, A) \in \mathbb{R}^{(h+2)\times h} \times \mathbb{R}^h$. We can represent the data samples as rows in a matrix $X \in \mathbb{R}^{n\times(h+2)}$—with $\mathbf{f}_i$ denoting the ith "feature" (i.e. column) of X—and the labels as elements of $\mathbf{y} \in \mathbb{R}^{n\times 1}$, as illustrated in Figure 4.

Choose k, l, m, n such that $k < l < m < n$ where $k > h$, $l - k > h$, $m - l > 2$ and $n - m > h$.

When $i \le l$, let

$$x_{i,j} = \begin{cases} i, & j = 1 \\ i-1, & j = 2 \\ 1, & i \equiv j \pmod h \\ -1, & i \not\equiv j \pmod h,\ i \le l \\ 0, & i \not\equiv j \pmod h,\ k < i \le l. \end{cases}$$

When $l < i \le m$, let

$$x_{i,j} = \begin{cases} -1, & j \le 2,\ i \equiv j \pmod 2 \\ 0, & j \le 2,\ i \not\equiv j \pmod 2 \\ 0, & j > 2,\ l < i \le m. \end{cases}$$